

APARNA KRISHNAN

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Education

Virginia Polytechnic and State University, Blacksburg, VA

August 2023 – Present

PhD. Theoretical Chemistry *Advisor : Prof T Daniel Crawford*

3.82/4.0

Honors: Phi Kappa Phi

Experience: Time-dependent coupled cluster, Linear response theory, Spectroscopy, Diagonalization algorithms, Analytic gradient theory, Numerical differentiation, Scientific automation pipelines

Birla Institute of Technology and Science Pilani, Goa

August 2018 – May 2023

Int. MSc.(Hons.) Chemistry and B.E.(Hons.) Mechanical Engineering

9.01/10.0

Honors: Rank 1, Chemistry

Coursework: Optimization, Group theory, Statistical Mechanics, Quantum Mechanics, Quantum Chemistry and Spectroscopy, Electronic Structure Theory, Finite Element Methods, Numerical Methods for Software Mathematics, Machine Learning

Experience

Virginia Polytechnic and State University

August 2022 – May 2023

Thesis Intern, Advisor: Prof T Daniel Crawford

Blacksburg, USA

- Automated the spin integration of coupled cluster equations and generation of code, reducing manual calculation time by 80 percent and allowing research team to focus on complex quantum chemistry challenges.
- Formulated a multi-scale approach, combining MD and DFT, and devised a Python script that automated MD frame sampling; enhanced Raman spectra generation, and solvent shell definition, leading to a significant reduction in runtime from 10 days to 12 hours.

Indian Institute of Technology, Bombay

May 2021 – March 2022

Research Intern, Advisor: Prof Rahul Maitra

Mumbai, India

- Established a rigorous theoretical understanding of CC theory and explored the applications in quantum chemistry.
- Developed an innovative engine that automates Wick's theorem. Also implemented a fully automatic equation and code generator for dual exponent formulation of CC theory involving scattering-type operators with minimal user input.

University of California, Berkeley

August – October 2020

Remote Research Intern, Advisor: Dr Ashok Ajoy

- Co-ordinated in the design and 3D modeling of Nitrogen Vacancy (NV) centers in nano-diamonds to study spin relaxation due to magnetic hyper-polarization.
- Integrated optical and MR imaging in diamond particles, naturally endowed with more than 1000 quantum defects, such as the NV centers, to open new avenues in the field of medicine and diagnosis like low-field MRI guided endoscopy.

Dhio Research and Engineering Pvt Ltd.

May – July 2020

Research Intern, Advisor: Santosh NL

Bangalore, India

- Simulated the visco-elastic properties of polymeric materials using J-OCTA, an advanced quantum simulation software and proposed the processing and performance of rubber type polymers in automotive industry.
- The project revealed critical polymer architecture that results in better on-road performance of tires. The calculations are 5x faster and accurate using ab-initio quantum mechanics.

Projects

Density matrix formulation of CISD vibrational circular dichroism (VCD)

- Developed a full pipeline for analytic VCD intensities within CISD using a density-matrix reformulation of atomic axial tensors (AATs), enabling efficient generalization to higher-level correlated methods.

Density-matrix formulation of MP2 vibrational circular dichroism (VCD)

- Extended the density-matrix AAT framework to MP2, implementing a complete VCD pipeline including analytic Hessian, normal-mode analysis, and rotational strengths.

GIAO extension to CISD-VCD (in progress)

- Incorporated gauge-including atomic orbitals (GIAOs) into the CISD-VCD framework to achieve origin-independent AATs, improving accuracy for basis-set-dependent magnetic response properties.

Coupled-cluster vibrational circular dichroism (CC-VCD) (in progress)

- Developing a CCSD-level VCD implementation building on the CISD density-matrix infrastructure, targeting a rigorous and efficient analytic derivative framework for chiral spectroscopy.

PyCC: Python-based coupled cluster package (contributor)

- Contributed to an open-source, reference implementation of coupled cluster methods (CCD, CC2, CCSD, CCSD(T)) with real-time and local-correlation extensions; implemented TD-EOM-CC methods, orbital decomposition analysis, and VCD pipelines within the framework.

Real-time coupled cluster methods for the calculation of field-dependent molecular properties.

- Revealed the orbital and symmetry contribution to peaks in an absorption spectra. Crafted an algorithm using autocorrelation as a sole metric for deriving field-dependent properties and orbital decomposition, which can reduce storage requirements by 95 percent.

A bi-orthogonal Jacobi-Davidson method to solve the eigenvectors of a non-Hermitian system.

- Engineered a novel bi-orthogonal Jacobi-Davidson (biJD) framework, accelerating eigenvector solutions for non-Hermitian systems by 25x compared to conventional methods, while also maintaining bi-orthogonality of solutions.

Mirror-assisted Digital Image Correlation (DIC) to extract side surface information.

- Designed and validated a mirror-assisted 3D-DIC technique for measuring side surface deformations in 3D printed carbon-reinforced PEEK, enhancing accuracy and reducing equipment complexity.

Skills

Quantum Computing: Qiskit, Cirq, quantum algorithms

Scientific Computing: Python, C++, Numpy, Scipy, Sympy, Pandas, h5py, Pickle, Matplotlib, einsum

Machine Learning: Pytorch, Tensorflow, Keras, NLP

Certifications: Quantum Computing (QWorld), Machine learning (Stanford), Quantum excellence (QGSS'25)

Achievements

KVPY Scholar, 2018 and 2019: Won the highly competitive scholarship given to students to pursue a career in Science by Department of Science and Technology, Govt of India twice, both in 2018 and 2019.

BITS Pilani Travel Award Winner: Awarded the prestigious BITS travel award to finish Bachelor's thesis in a foreign university.

Elected member of Phi Kappa Phi: Elected to Phi Kappa Phi for outstanding academic standing at Virginia Tech.

Outstanding Poster Award: Won the best poster award at SETCA'26 in Atlanta.

Conferences presented

SETCA'23, USC, Columbia:Determination of Optical Properties by Modeling Complex Solvent Effects : Molecular Dynamics and Density Functional Approach

ESQC, Palermo, Italy:Understanding Real-Time Coupled-Cluster Absorption Spectra: Orbital Contributions from Individual Transitions

SERMACs, Atlanta, Georgia:Understanding Real-Time Coupled-Cluster Absorption Spectra: Orbital Contributions from Individual Transitions

SETCA'25, UM, Tennessee:Orbital-Resolved Absorption Spectra from RT-CC Autocorrelation function

MRPSS'25, Stockholm, Sweden:A tale of two decompositions: Analyzing electronic excitations in real-time coupled-cluster

SETCA'26, GSU, Atlanta:Density-matrix formulation of atomic axial tensors for CISD vibrational circular dichroism

Manuscripts in preparation

1. Chemical Interpretation of Time-Dependent Coupled-Cluster Theory, arXiv:2605.17409
2. Field-Strength Dependence and Fourier Transform Methods in Autocorrelation Function Decomposition, in preparation
3. Deformation Characterization of Additive Manufactured Polyether Ether Ketone Using Mirror Assisted Image Correlation, submitted
4. Density-Matrix Formulation of Atomic Axial Tensors for Vibrational Circular Dichroism within Configuration Interaction Theory, submitted
5. Density-Matrix Formulation of MP2 VCD, in preparation
6. PyCC: An opensource Python-based coupled cluster package, in preparation